

Assignment 6:

Objective:

- Divide given networks into **subnets**
- Assign IPs to devices based on subnetting
- Connect **multiple subnetted networks using routers**
- Enable communication using **static routing**

PC 2 config:

IP: 172.16.2.10

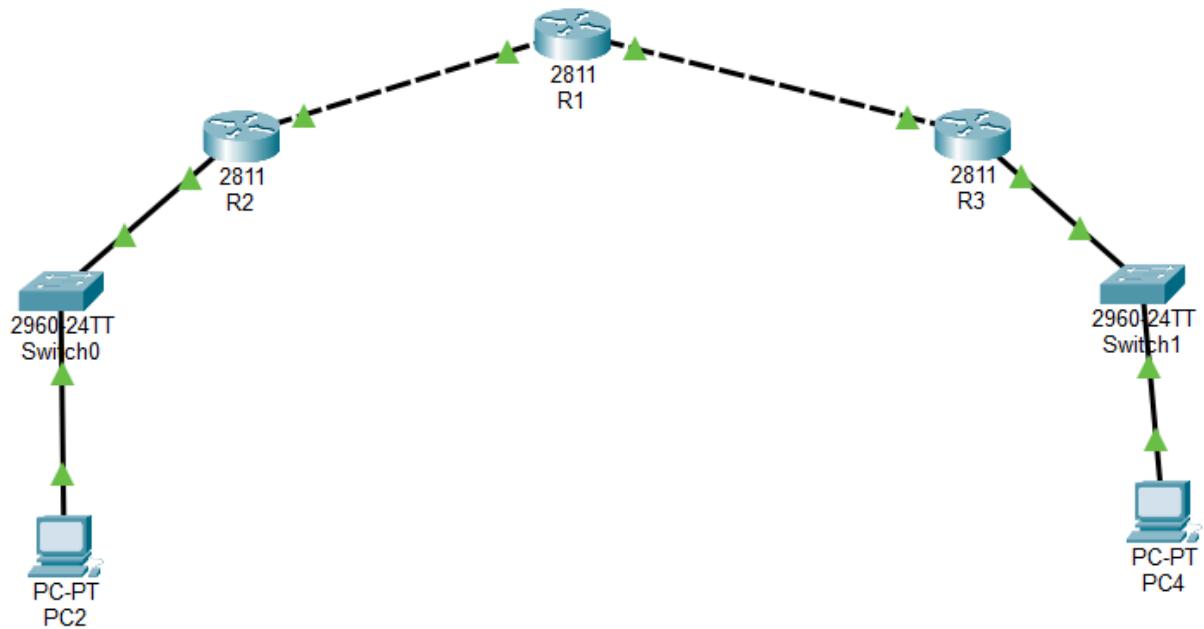
Gateway: 172.16.2.1

PC 4 config:

IP: 10.2.2.20

Gateway: 10.2.2.1

NETWORK:



CONFIGURATION:

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int G0/0/0
Router(config-if)#ip add 100.1.1.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

Router(config-if)#int G0/0/1
Router(config-if)#ip add 20.1.1.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

Router(config-if)#ex
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int Gig0/0/1
Router(config-if)#ip add 172.16.2.1 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
```

```
Router(config-if)#ex
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto summary
Router(config-router)#network 20.1.1.0
Router(config-router)#network 10.2.2.0
Router(config-router)#ex
Router(config)#ex
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.2.2.0/24 is directly connected, GigabitEthernet0/0/1
L       10.2.2.1/32 is directly connected, GigabitEthernet0/0/1
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.1.1.0/24 is directly connected, GigabitEthernet0/0/0
L       20.1.1.2/32 is directly connected, GigabitEthernet0/0/0
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up

Router(config-if)#int Gig0/0/0
Router(config-if)#ip add 100.1.1.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

Router(config-if)#ex
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto summary
Router(config-router)#network 100.1.1.0
Router(config-router)#network 20.1.1.0
Router(config-router)#ex
Router(config)#ex
Router(config-if)#int Gig0/0/1
Router(config-if)#ip add 10.2.2.1 255.255.255.0
Router(config-if)#no shut
Router(config-if)#ex
Router(config)#int Gig0/0/0
Router(config-if)#ip add 20.1.1.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up
```

CONNECTION VALIDATION:

Cisco Packet Tracer PC Command Line 1.0

C:\>ping 172.16.1.10

Pinging 172.16.1.10 with 32 bytes of data:

Reply from 172.16.1.10: bytes=32 time=8ms TTL=128

Reply from 172.16.1.10: bytes=32 time=7ms TTL=128

Reply from 172.16.1.10: bytes=32 time=7ms TTL=128

Reply from 172.16.1.10: bytes=32 time=6ms TTL=128

Ping statistics for 172.16.1.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 6ms, Maximum = 8ms, Average = 7ms

C:\>ping 10.2.2.20

Pinging 10.2.2.20 with 32 bytes of data:

Reply from 172.16.2.1: Destination host unreachable.

Reply from 172.16.2.1: Destination host unreachable.

Request timed out.

Reply from 172.16.2.1: Destination host unreachable.

Ping statistics for 10.2.2.20:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),